MIL-F-18280A(ASG)
24 APRIL 1956
Superseding
MIL-F-18280(Aer)
5 November 1954

MILITARY SPECIFICATION

FITTINGS, FLARELESS, FLUID CONNECTION

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 Scope.- This specification covers flareless, fluid connection fittings for use in aircraft.

1.2 Classification.- Fittings shall be furnished in the types and styles designated by the applicable MS Standards referred to hereinafter, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
QQ-A-266	Aluminum Alloy 14S; Bars, Rods, Shapes, and Wire-Rolled or Drawn
QQ-A-267	Aluminum Alloy 24S; Bars, Rods, and Shapes-Extruded
QQ-A-268	Aluminum Alloy 24S; Bars, Rods, and Wire-Rolled or Drawn
QQ-A-355	Aluminum-Alloy (24S), Plate and Sheet
QQ-A-367	Aluminum-Alloy, Forging, Heat-Treated
QQ-M-151	Metals; General Specification for Inspection of
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-624	Steel, Alloy; Bars (General Purpose)
QQ-S-633	Steel, Carbon; Bars (General Purpose)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-676	Boxes, Set-Up, Paperboard

Military

JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible

FSC 4730

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MIL-P-116	Preservation, Methods of
MIL-B-4229	Boxes; Paperboard, Metal-Stayed
MIL-O-5606	Oil; Hydraulic, Aircraft, Petroleum Base
MIL-S-5626	Steel; Chrome-Molybdenum (4140) Bars, Rods, and Forging Stock (for Aircraft Applications)
MIL-S-6049	Steel; Chrome-Nickel-Molybdenum (8740) Bars, Rods, and Forging Stock (for Aircraft Applications)
MIL-S-6050	Steel; Chrome-Nickel-Molybdenum (8630) Bars, Rods, and Forging Stock (for Aircraft Applications)
MIL-H-6088	Heat Treatment of Aluminum Alloys; Process for (Aircraft Application)
MIL-S-6758	Steel; Chrome-Molybdenum (4130) Bar, Rod, and Forging Stock (for Aircraft Applications)
MIL-T-6845	Tubing; Steel, Corrosion-Resisting, (18 - 8) Aircraft, Hydraulic System
MIL-H-6875	Heat Treatment of Steels (Aircraft) Process for
MIL-T-7081	Tubing, Aluminum Alloy (61S), Seamless, Round, Hydraulic System
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-T-8504	Tubing, Steel, Corrosion-Resistant (18 - 8) Annealed, Aircraft Hydraulic System
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8695	Steel, Bars and Wire, Molybdenum (4037) (Aircraft Quality)
MIL-S-13433	Steel, Leaded; Bars

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

For applicable MS Standards for fittings covered by this specification, see Supplement 1.

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN815	Union - Flared Tube
AN818	Nut, Coupling
AN819	Sleeve - Coupling
AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND10061	Tubing End - Standard Dimensions for Flared
AND10064	Fittings - Installation of Flared Tube, Straight Threaded Connectors
AND10078	Tubing End - Standard Dimensions for Double Flare

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 425	MS Fittings, Classification of Defects of

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The fittings furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials.- Fittings shall be fabricated of materials listed in table I, as specified on the applicable MS Standard.

3.2.1 Steel.- All steel shall be made by the open-hearth or electric-furnace process.

TABLE I

Materials

Material		Specification No.
Aluminum alloy -	Bars, shapes, sheet, strip, and plate	QQ-A-266, Cond T6 QQ-A-267, Cond T6 QQ-A-268, Cond T6 QQ-A-355, Cond T6
	Forgings	QQ-A-367, Class 5, Cond T6
Steel -	Bars, rods, and forgings:	QQ-S-624, FS1335
	Nuts, connectors, and component fittings	MIL-S-6049 MIL-S-6758 MIL-S-8695 MIL-S-6050 QQ-S-633, FS1137, or FS1141
	Sleeves	MIL-S-13433 C-1212 C-1010 C-1113 C-1213 C-1112
Corrosion-resistant steel - Bars, rods, and forgings:		
	Nuts, connectors, and component fittings	Type 302 303 304 304L 316 321 347
	Sleeves	Type 17-4-PH

3.2.2 Heat treatment.-

3.2.2.1 Aluminum alloy.- Aluminum-alloy materials shall be heat-treated in accordance with Specification MIL-H-6088.

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3.2.2.2 Steel.- When required, steel shall be heat-treated in accordance with Specification MIL-H-6875.

3.2.2.3 Standard MS21922 sleeves shall be heat-treated as specified in 4.4.3.2.

3.2.3 Physical properties.- Unless otherwise specified on the applicable MS Standard, the hardness of the steel parts, excluding sleeves, shall be Rockwell B92-C31.

3.2.4 Alternate material.- Materials not listed in table I and use of which has not been specifically prohibited may be used only upon specific approval of the procuring activity. In order to obtain this approval, it will be necessary to conduct complete performance tests for alternate materials and submit a test report as specified herein. In no case shall the alternate material be used in production fittings prior to receipt of specific approval from the procuring activity.

3.2.5 Except when specifically specified on the MS Standard, castings of any type shall not be used.

3.2.6 Fittings made by welding or brazing together two or more separate pieces of metal may be used on Service aircraft only when the design method of fabrication by each manufacturer offering such fittings has been submitted and approved.

3.3 Design and dimensions.- The design and dimensions of MS Standard fittings shall be in accordance with the applicable MS Standard.

3.3.1 Shape and form restrictions. Abrupt reductions of section shall be avoided. Small external sections adjoining relatively heavy body sections shall be shaded into the heavier sections by means of ample fillets. Sharp corners or inadequate fillets, excessive undercuts, or excessive grooves at the junction of such small sections with large sections of fittings shall be considered cause for rejection. Unless otherwise specified, drawing tolerances shall apply to fillet and corner radii.

3.3.2 Passages.-

3.3.2.1 Drill offset.- On straight fittings where the fluid passage is drilled from each end, the offset between the drilled holes at the meeting point of the drills shall not exceed 0.015 inch. It shall be possible to pass through the fluid passage a ball whose minimum diameter is 0.020 less than the minimum diameter specified for the passage. This does not mean that the drilled passage may be smaller than that required by the MS Standard.

3.3.2.2 On angle fittings, the cross-sectional area at the junction of the fluid passages shall be not smaller than the cross-sectional area of the smaller passage.

3.3.2.3 Wall thickness.- Except as otherwise specified on the applicable MS Standard, the wall thickness at any point on the fitting shall be not less than the thickness established by the dimensions and tolerances for the inside and outside diameters and eccentricities specified in the MS Standard.

3.3.2.4 Threads.- Except as otherwise specified, all straight threads shall be unified form and shall conform to Specification MIL-S-7742.

3.4 Finish.- All fittings shall be finished as noted in table II in accordance with the applicable finish specification, and as specified in 3.4.1 through 3.4.1.6, inclusive.

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TABLE II

Finish

Fitting material	Type of finish	Finish specification
Aluminum alloy	Anodic Treatment	MIL-A-8625, type II
Steel	Cadmium plate	QQ-P-416, type I, class C
Corrosion-resistant steel	Passivation	-- -- --

3.4.1 Anodic treatment of aluminum-alloy flareless fittings.- Supplementing Specification MIL-A-8625, aluminum-alloy fittings incorporating ends conforming to Standards MS33514 and MS33515 shall be subject to the following requirements.

3.4.1.1 Etching.- Aluminum-alloy fittings shall not be etched after machining.

3.4.1.2 Sulphuric-acid concentration.- This shall be 15 percent to 18 percent by volume.

3.4.1.3 Current density.- Anodic current density shall be 12 to 16 amperes per square foot when measured on a flat sheet.

3.4.1.4 Time. Aluminum-alloy fittings shall be anodized for 30 ±5 minutes.

3.4.1.5 Continuity of coating.- The anodic coating of internal surfaces, ends, threads, and thread reliefs, shall be clean, unbroken, and free from rack-contact marks. External rack-contact marks shall not be cause for rejection.

3.4.1.6 Sealing.- Fittings shall be sealed in 1 percent to 5 percent potassium dichromate solution by weight, for 10 ±2 minutes at 185° ±5°F.

3.5 Identification of product.- All MS Standard tube fittings shall be marked in accordance with the following instructions. The marking shall be applied in a location not detrimental to the fittings and shall not be detrimental to the corrosion protection of the fitting.

3.5.1 MS Standard symbols and trade-marks.- Unless otherwise specified, all fittings, except sleeves, shall be marked with the letters "MS" and manufacturer's name or trade-mark. Standard MS21922 sleeves shall be permanently trade-marked only. On fittings other than sleeves, the marking shall be permanent, consisting of embossed letters or impression stamping. The letters "MS" shall be separated and distinct from the manufacturer's name or trade-mark.

3.5.2 Identifying number or letter.-

3.5.2.1 Wherever practicable, each fitting shall also be identified by the complete drawing number as shown by the MS Standard. The size is not to be included with the MS Standard No. The identifying number shall be added to the part by embossing, impression stamping, ink stamping, or shall be printed on a paper collar supplied with the fitting. The preferable marking shall be in the order mentioned. The identifying number shall not be placed on Standards MS21914, MS21918, and MS21922. Also, the fittings in sizes -6 and under need not be individually marked but shall be packaged in units, with the package carrying the identification number.

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3.5.2.2 Each MS Standard fitting, except the Standard MS21922 sleeve, made from corrosion-resistant steel shall be permanently marked with the letter "S" if fabricated from either type 304L or 347 corrosion-resistant steel, and with the letter "C" if fabricated from either 302, 303, 304, 316, or 321 corrosion-resistant steel.

3.5.3 Color identification.- In addition to the marking specified under 3.5.1, 3.5.2.1, and 3.5.2.2, the MS Standard fittings shall also be identified by the following color or finish:

Aluminum alloy	No dye (See 3.5.3.1)
Steel fittings	Natural cadmium plate
Steel sleeves	Natural cadmium plate
Corrosion-resistant steel	Natural finish

3.5.3.1 Color.- Aluminum-alloy fittings shall not be artificially colored by dyeing but may be yellow in color owing to dichromate seal.

3.6 Workmanship.- Fittings shall be free from cracks, laps, seams, burrs, longitudinal and spiral tool marks, or any other defects which may detrimentally affect their suitability for the service intended. Contractor's production and inspection processes shall be established so the end results will produce a completed article in strict compliance with specification and contract requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of flareless fittings shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.
(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 General testing practice for Qualification tests.-

4.2.1 Thread lubricant.- No thread lubricant shall be used on straight thread fittings other than hydraulic fluid conforming to Specification MIL-O-5606 unless specifically approved by the procuring activity.

4.2.2 Tightening values.- The normal tightening values of the fittings are minimum of 1/6 turn, and maximum of 1/3 turn past the point of sharp torque rise for all sizes and all types of tube and fitting materials. Torque values for "overtightened" assembly conditions for all sizes shall be as shown in table III. These values apply to resetting operations only, as specified on Standard MS33566.

4.2.3 Tube material and working pressures.- Fittings shall be tested with the tube materials specified in table III. The operating pressures listed in table III shall be considered normal working pressures for high-pressure fittings.

4.2.4 Test fluid.- Fluid conforming to Specification MIL-O-5606 shall be used in all tests.

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TABLE III

Tube material and pressures

Tube OD (nominal) size		Aluminum alloy 61S-T-6 (Specification MIL-T-7081) condition T			Stainless steel 18-8 (Specification MIL-T-6845) 1/4 hard			Stainless steel 18-8 (Specification MIL-T-8504) annealed		
		Wall	Work press.	Over-tightened torque (in.lb)	Wall	Work press.	Over-tightened torque (in.lb)	Wall	Work press.	Over-tightened torque (in.lb)
1/8	2	0.020	3,000	100	0.016	3,000	100	0.020	3,000	100
3/16	3	.028	3,000	120	.018	3,000	120	.020	3,000	120
1/4	4	.035	3,000	140	.020	3,000	140	.028	3,000	140
5/16	5	.049	3,000	170	.022	3,000	180	.028	3,000	180
3/8	6	.049	3,000	220	.025	3,000	220	.035	3,000	220
1/2	8	.065	3,000	440	.028	3,000	540	.049	3,000	540
5/8	10	.083	3,000	800	.035	3,000	1,100	.058	3,000	1,100
3/4	12	.095	3,000	1,200	.042	3,000	1,750	.065	3,000	1,750
1	16	.065	1,500	1,950	.065	3,000	3,100	.083	3,000	3,100
1-1/4	20	.083	1,500	2,100	.065	1,500	3,700	.065	1,500	3,700
1-1/2	24	.095	1,500	2,300	.065	1,500	4,050	.065	1,500	4,050
2	32	.156	1,500	2,500	.065	1,500	4,400	.095	1,500	4,400

4.3 Qualification tests.-

4.3.1 Test samples.- For each size upon which qualification is desired, specimens as indicated in table IV shall be submitted unassembled to the Service designated in 6.3, plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification tests
FITTING, FLARELESS, FLUID CONNECTION
MS Part No.
Name of manufacturer
Manufacturer's Part No.
Submitted by (name) (date) for Qualification tests in accordance with the requirements of Specification MIL-F-18280A under authorization (reference authorizing letter).

4.3.2 The purpose of the Qualification tests is essentially to qualify the joint. Qualification of the assemblies called out in table IV for any given size, will automatically qualify the fittings of that size of the same part numbers tested, but of different materials. Qualification of the assemblies called out in table IV for any given size, will also automatically qualify the other types of fittings of the size tested.

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TABLE IV

Qualification test samples

Test	Tubing conforming to Specification MIL-T-6845, 1/4 hard		Fittings consisting of MS21921 Al nut, MS21922 steel sleeve, and MS21902 Al union
	No. of specimens	Length (inches)	
4.3.3.1 Repeated assembly			
4.3.3.1.1 Minimum tightening values	2	12	4
4.3.3.1.2 Maximum tightening values	2	12	4
4.3.3.1.3 Overtightened torque	2	12	4
4.3.3.2 Impulse			
Room temperature, 175,000 cycles	2	12	4
Low temperature, 25,000 cycles	2	12	4
4.3.3.3 Vibration - Impulse 1/	12	16 2/	12
Extra specimens	6	16	12
TOTAL	(10 (18	12 16	44

1/ Test specimens not required for 16-inch size and above on Vibration - Impulse test.

2/ Six lengths of the 16-inch tubing are for use in assembly of AN fittings for the Vibration - Impulse test.

4.3.3 Tests.- Qualification tests shall consist of all the tests of this specification as described under "Acceptance tests" and, in addition, each size fitting shall be subjected to the following tests.

4.3.3.1 Repeated assembly tests.-

4.3.3.1.1 Minimum tightening values.- The fitting assemblies shall be tightened 15 successive times, using the minimum tightening values specified in 4.2.2. Each tightening operation shall include a complete removal of the tube from the fitting body. After each third tightening operation, the fitting assemblies shall be subjected to fluid pressure of 2 times the working pressure (held for 5 minutes). There shall be no leakage of the fitting assemblies when subjected to these pressures. At no time during the 15 tightening operations shall the assemblies be difficult to assemble or disassemble. After the 15th tightening operation, there shall be no leakage or blowoff of the fitting assemblies up to the burst pressure of the tubing or to 4 times the working pressure, (held for 5 minutes), whichever is the lower pressure.

4.3.3.1.2 Maximum tightening values.- The tests specified under "Minimum tightening values" shall be repeated with the same results except that the fitting assemblies shall be tightened to the maximum tightening values specified in 4.2.2.

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4.3.3.1.3 Overtightened torque.- The tests specified in 4.3.3.1.1 shall be repeated with the exception that the fitting assemblies shall be tightened to the overtightening torque values listed in table III. The assemblies shall not leak or blowoff when the test pressures are applied, but moderate difficulty in assembly and disassembly shall not be cause for rejection.

4.3.3.1.4 Torque on tubing during assembly.- During the first tightening to minimum tightening values, the tendency for the tube to turn during assembly shall be recorded and shall not exceed the values given in table V.

TABLE V

Turning torque on tube
Inch-pounds (max)

Tube size	2	3	4	5	6	8	10	12	16	20	24	32
Torque	5	5	5	5	10	15	20	20	25	25	25	25

4.3.3.1.5 Minimum opening.- At no time shall any of the assemblies tested under "Minimum"and"Maximum"tightening values present an internal diameter opening through either fitting or tube less than those specified in table VI.

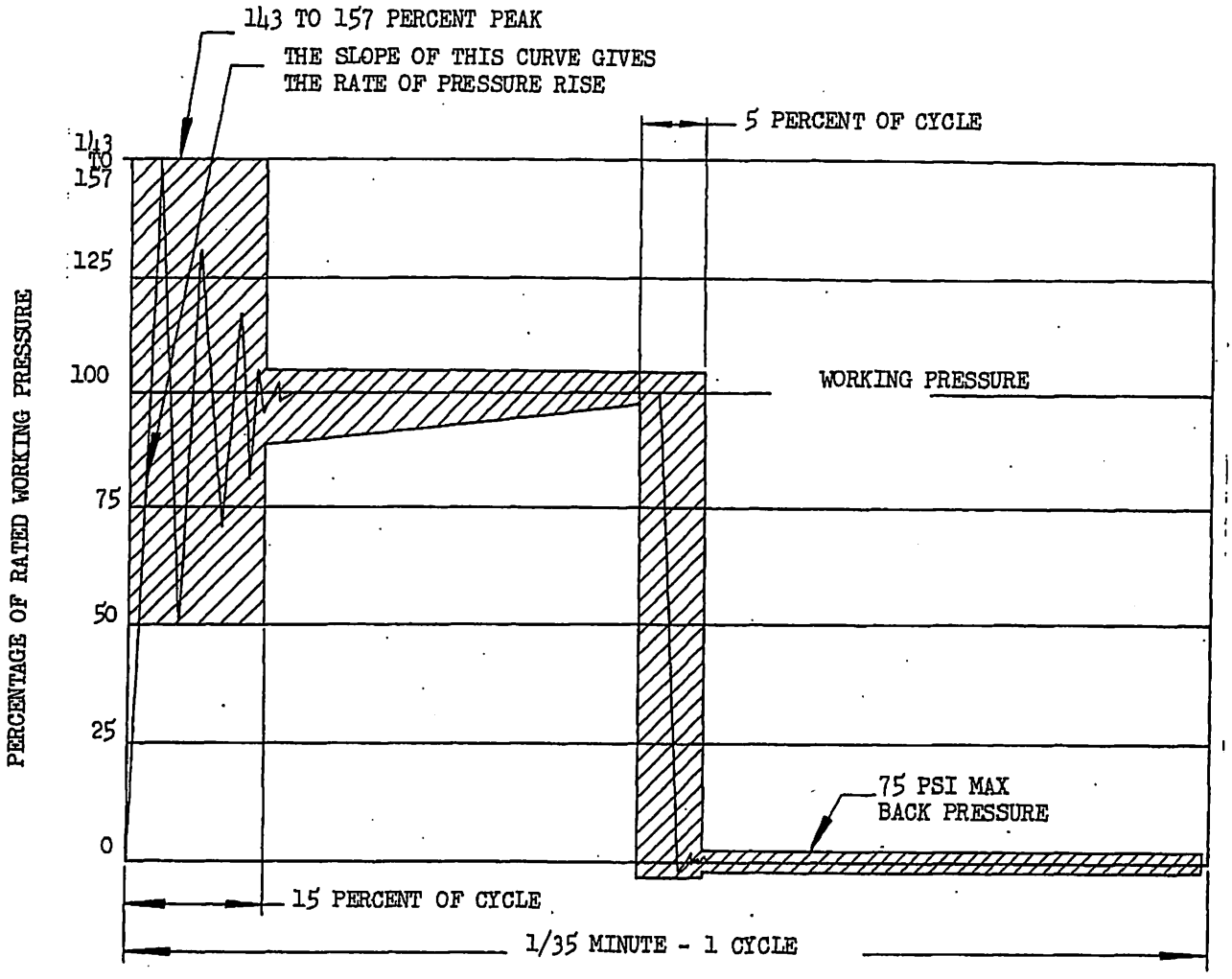
TABLE VI

Minimum opening in inches when using tubing specified in table IV

Tube Size	Specification		
	MIL-T-7081	MIL-T-6845	MIL-T-8504
2	0.060	0.070	0.060
3	.095	.110	.115
4	.150	.165	.155
5	.180	.225	.225
6	.240	.290	.270
8	.330	.400	.380
10	.420	.485	.475
12	.530	.610	.590
16	.830	.840	.800
20	1.050	1.080	1.080
24	1.280	1.330	1.330
32	1.650	1.780	1.780

4.3.3.2 Impulse test.- All impulse tests shall be conducted at a rate of 35 ±5 cpm. Each impulse cycle shall constitute a rise from 0 psi to surge pressure and drop to zero as shown in figure 1. Hydraulic fluid shall be used as the testing medium with a peak surge pressure of 1.43 to 1.57 times the working pressure, as shown by an oscilliscope. Tubing assemblies at each of the tightening values noted below shall be tested at room temperature for 175,000 impulse cycles, followed by 25,000 impulse cycles, at -54°C (-65°F). The assemblies shall be subjected to 2- to 10-psi static pressure for 24 hours at -54°C (-65°F) prior to the low temperature impulse cycle. During this cold-soak period, there shall be no leakage. Before and after the impulse cycles at both room temperature and -54°C (-65°F), the fitting assemblies shall be tested at proof pressure for 5 minutes, during which period there shall be no fluid leakage or fitting failure.

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THE CURVE SHOWN ABOVE IS THE APPROXIMATE PRESSURE-TIME CYCLE DETERMINED TO BE OF PROPER SEVERITY FOR IMPULSE TESTING OF FLARELESS FITTINGS. WHILE IT IS MANDATORY ONLY THAT PRESSURE PEAK RISES TO 143 TO 157 PERCENT OF THE WORKING PRESSURE AT SOME POINT PRIOR TO LEVELING OFF AT RATED PRESSURE, IT IS CONSIDERED HIGHLY DESIRABLE THAT THE PRESSURE-TIME CURVE BE CONFINED TO THE SHADED AREA INDICATED. ONE VERY DESIRABLE BENEFIT TO BE GAINED IN THIS MANNER IS THAT RESULTS OF TESTS PERFORMED ON DIFFERENT TEST MACHINES WILL BE MORE NEARLY COMPARABLE.

FIGURE 1. Dynamic pressure impulses

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4.3.3.2.1 At minimum tightening values.- Assemblies shall be tightened to applicable minimum tightening values and subjected to the Impulse test. There shall be no leakage or failure of any of the test assemblies.

4.3.3.2.2 At overtightened torque.- Assemblies shall be tightened to the overtightened wrench torque as specified in table III, and subjected to the Impulse test. There shall be no leakage or failure of any of the test assemblies.

4.3.3.3 Vibration - Impulse test.- The vibration test shall consist of mounting a fitting and tube assembly, as noted below, in a vibration machine, as specified below, and stressing the tube, in a linear motion in a plane through the tube axis, with respect to the fixed end of the fitting. The ability of the test fitting to withstand vibration shall be compared to that of an AN Standard fitting.

4.3.3.3.1 Fitting installation.- For installation of the AN fitting, an AN815 type fitting with AN818 nut and AN819 sleeve shall be assembled on tubing flared in accordance with Drawing AND10061 or AND10078, using the minimum wrench torques specified in Drawing AND10064. Flareless tube fittings being tested shall be assembled with the minimum tightening value of 1/6 turn recommended for the fitting. All fittings shall be mounted into a boss in accordance with Drawing AND10050 on the vibration machine.

4.3.3.3.2 Vibration machine.- The vibration machine shall be so designed that the tube assemblies with fittings at each end can be mounted as follows:

(a) One end shall be rigidly mounted.
(b) The other end shall be so mounted that it will be free to slide in one plane.

The machine shall have provisions to displace one of the ends from the axial line through fitting and tube, thus stressing the specimen at a speed of 1,750 ±50 cpm. It shall be possible to apply impulse pressure from 0 psi up to the peak surge pressure of 1.43 to 1.57 times the working pressure at a rate of 35 ±5 cpm, to the fitting and tube assembly during vibration.

4.3.3.3.3 Conducting Vibration - Impulse tests.- The fitting and tube assembly shall be mounted in the vibration machine and the displacement and length of assembly shown in figure 2, shall be such as to stress the tubing to the values listed in table VII. The tube length shall be calculated as a cantilever beam with a concentrated load on the free end. The formula to be used is as follows:

$$l = \sqrt{\frac{3DEc}{S}} \qquad (1)$$

l = length as shown in figure 2, in inches
D = displacement in inches
E = modulus of elasticity (30,000,000)
c = 1/2 tube diameter in inches
S = stress of tube material (see table VII)

The above formula was derived by the combination of the following formulas:

$$S = \frac{Mc}{I} = \frac{Flc}{I} \qquad (2)$$

$$D = 1/3 \frac{Fl}{EI}^3 \qquad (3)$$

The specimen shall be subjected to vibration while impulse pressures, as described in 4.3.3.2, are being simultaneously applied until failure occurs, or until completion of 10,000,000 vibration cycles, (approximately 100 hours). The average number of cycles

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before failure for each size of flareless-tube fitting shall be the same or more than the average number of cycles withstood by the corresponding size of AN tube fitting. Not more than one specimen of each size of a flareless-tube fitting shall have a lower life than the lowest vibration life of the corresponding AN fitting. When comparing vibration results, AN and flareless, the inferior fitting shall have a minimum average life of 100,000 vibration cycles.

TABLE VII

Stress of tube material
Conforming to Specification MIL-T-6845

Size	3	4	5	6	8	10	12
Psi	40,000	40,000	40,000	40,000	40,000	25,000	20,000

4.3.3.4 Additional tests.- Additional tests may be performed by the Services if considered necessary.

4.4 Acceptance tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the required tests.

4.4.1 Material certification.- Certificates showing conformance with the applicable material specification shall be available to the Inspector for each lot of fittings.

4.4.2 Examination of product.- All fittings shall be examined to determine conformance with this specification with respect to the requirements of workmanship, marking, surface defects, and finish.

4.4.3 Sampling.-

4.4.3.1 Threads, finish, and dimensions.- Sampling for threads, finish, and dimensions shall be random in accordance with Standard MIL-STD-105, and ANA Bulletin No. 425.

4.4.3.2 Sleeves, heat treatment (steel sleeves only).- A minimum of 3 pieces and a maximum of 10 pieces shall be inspected from each lot. A lot shall consist of finished parts which are of the same size, fabricated by the same process, heat-treated in the same lot, and submitted for inspection at the same time.

4.4.3.2.1 Procedure.- The following test shall be performed. Remove plating, grind and polish end of sleeve perpendicular to sleeve axis. In this manner the pilot or lip is removed. Polish the sleeve on No. 1 emery paper making sure that the scratches have been removed before continuing on through with the next No. 0, No. 2/0, No. 3/0 emery papers. Immerse samples in etchant (2 percent Nital) for a sufficient length of time for the desired etch, about 8 to 10 seconds. Wash in hot water, immerse in alcohol, and blow dry. Examine specimen with 10.25 objective and Filar eyepiece. Take at least three measurements at different points on the sleeve. At least one reading should be near the ID and one near the OD. Samples shall show a case depth of 0.0015 to 0.0019 inch when converted from Filar readings to inches, using standard calibrated conversion tables. When measured on a superficial hardness tester, the hardness of the core shall not exceed Rockwell B-99. The hardness of the case may be measured on a diamond point Tukon hardness tester and shall be not less than Rockwell C-50. Any variation from the above hardness values shall be sufficient cause for rejection.

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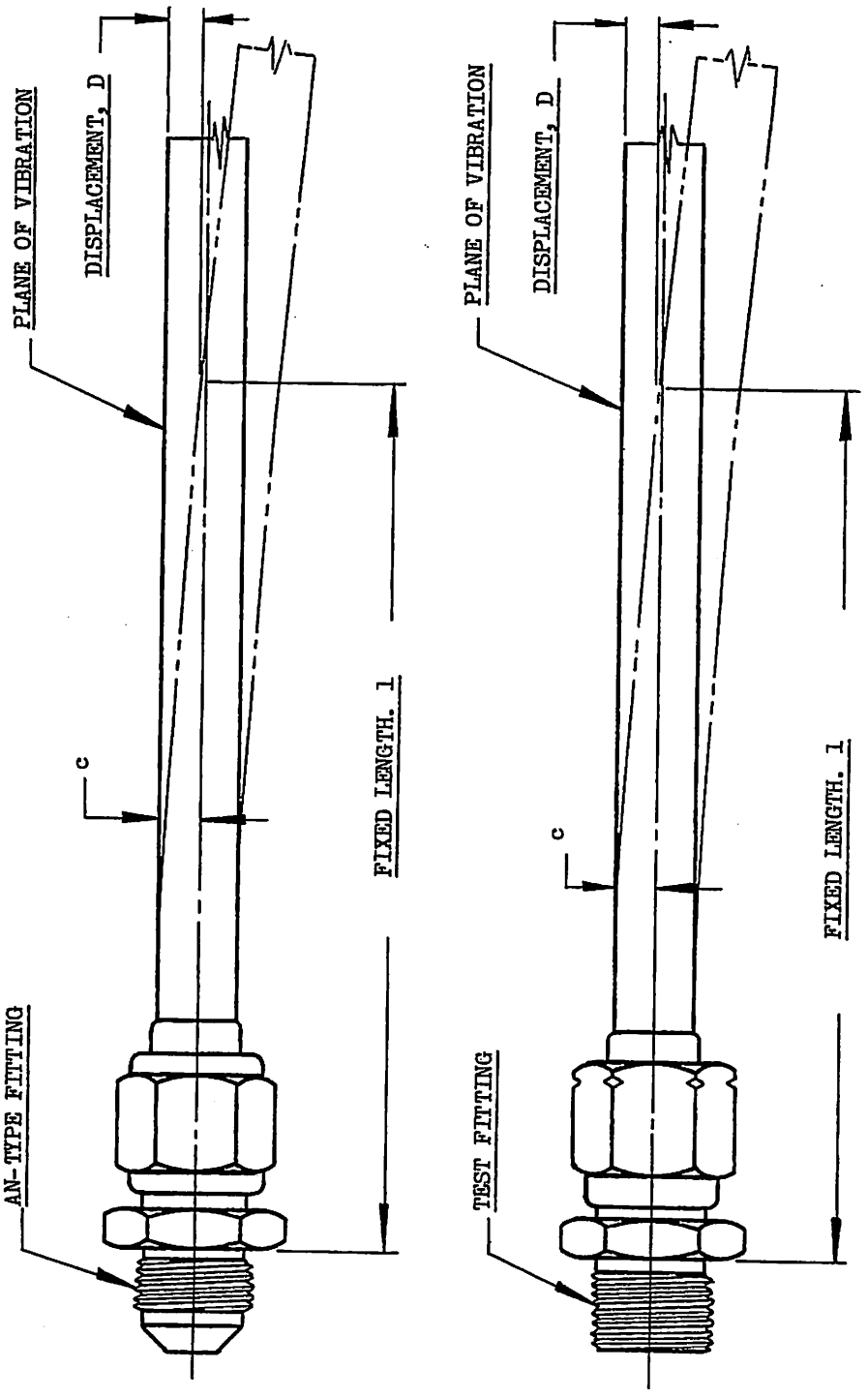


FIGURE 2. Mounting of fitting and tube assembly for Vibration - Impulse test

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4.4.3.3 Sleeves - Cut testing.- A minimum of 3 pieces, or 1/4 percent, whichever is greater, and a maximum of 1 percent shall be inspected from each lot. A lot shall consist of all finished parts which are of the same size, fabricated by the same process, heat-treated in the same lot, and submitted for inspection at the same time. The sleeves shall be assembled to Specification MIL-T-6845 corrosion-resistant tubing adhering to standard assembly procedure. (Hand preset 1-1/6 turns past sleeve grip.) The sleeve shall be driven back to expose the ring cut for examination. The cut shall encircle by 90 percent the periphery of the tube. The tube collapse shall not be excessive. (See table VI.) The bulge of the sleeve through the barrel shall not be excessive. There shall be no longitudinal or circumferential cracks on the sleeve before driving back. Circumferential crazing marks in the neighborhood of the pilot shall be allowed.

4.4.4 The Government Inspector may require magnetic inspection or other inspection tests whenever Government Service experience warrants.

4.4.5 Rejection and retest.-

4.4.5.1 If any sample quantity of a given lot fails to conform to the requirements of the sampling tables specified herein for threads, finish, and dimensions, 4.4.3.1, the entire lot shall be rejected or, at the discretion of the contractor, the entire lot shall be inspected for the test property in which it failed and all defective components rejected. If any sample of a given lot fails to conform to the acceptance sampling requirements of any of the remaining tests, the entire lot shall be rejected.

4.4.5.2 If one sample fails to pass the test specified in 4.4.3.2, a retest shall be necessary involving twice the number of samples. One defective sample in the retest shall be cause for lot rejection.

4.4.5.3 If one sample fails to pass the test specified in 4.4.3.3, a retest shall be necessary involving twice the number of samples. One defective sample in the retest shall be cause for lot rejection.

5. PREPARATION FOR DELIVERY

5.1 Application:- The requirements of section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and packaging.- Unless otherwise specified, fittings shall be protected from corrosion by packaging in accordance with Method 1 (c) of Specification MIL-P-116, without the use of a contact preservative. The quantity per unit and intermediate container shall be as specified below:

Size (inches)	Number per unit container	Number per intermediates
1/8 through 3/8	5	20
1/2 through 1	5	10
1-1/4 through 1-1/2	5	0
2	1	0

5.2.1 Unit containers.- Fittings up to and including 1 inch shall be packaged in commercial-type paperboard or fiberboard cartons. Each such carton shall have interior separation capable of supporting and separating each individual fitting in such manner as to prevent movement or contact of the fittings, one with the other. Fittings over 1 inch shall be packaged in containers conforming to Specification PPP-B-566, PPP-B-676, or MIL-B-4229.

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5.2.2 Nuts and sleeves.-

Size (inches)	Number per unit container
1/8 through 1/2	100
5/8 through 1-1/4	25
1-1/2 through 2	5

5.2.2.1 Unit containers.- Nuts and sleeves of one type and size shall be packaged in the following containers: Slip cover, friction top or hermetically sealed cans, or cartons conforming to Specification PPP-B-566, PPP-B-676, or MIL-B-4229.

5.3 Packing.- Unless otherwise specified, all items shall receive domestic packing. Shipping containers shall contain the same number of unit packages, shall be uniform in size, and snugly packed. The gross weight of the fully packed container shall not exceed approximately 200 pounds.

5.3.1 Domestic shipment.- Unless otherwise specified, interior packages shall be overpacked in substantial commercial exterior shipping containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. The use of corrugated or solid fiberboard having a minimum Mullen test of less than 275 pounds is prohibited. Containers shall be able to withstand storage, re-handling, and reshipment without the necessity of repacking.

5.3.2 Overseas shipment.- Unless otherwise specified, for overseas shipment, the interior containers shall be packed in boxes in accordance with Specification PPP-B-601 or JAN-P-106. Plywood, if used, shall be in accordance with Specification NN-P-515, type II, class 2. A sealed case liner conforming to Specification JAN-P-125 is required.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified
in the purchase document*
FITTING, FLARELESS, FLUID CONNECTION
Specification MIL-F-18280A
Manufacturer's Part No.

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification as provided by the procuring activity.

6. NOTES

6.1 Intended use.- Flareless fittings covered by this specification are intended for use in aircraft.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and style of fitting required. (See 1.2.)

(c) Whether domestic or overseas packing is required. (See 5.3.)

6.2.1 Qualified Products List.- Fittings considered acceptable under this specification are listed in the latest revision of QPL-18280.

6.3 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., the activity responsible for qualification, with a copy to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.

6.3.1 It is to be understood that fittings supplied under contract will be identical in every respect to the Qualification test samples tested and found satisfactory and to the detailed drawings thereof, except for changes previously approved by the activity responsible for qualification. Any unapproved changes from the Qualification test samples or from the MS Standards will constitute cause for rejection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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